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COMMON QUARTERLY EXAMINATION - 2026

Standard - 12

Reg.No. | | | | | | |

Time Allowed:3.00 Hours **PHYSICS** *Maximum Marks:70*

15×1=15

I. Answer all the questions:

1) Two identical conducting balls having positive charges q_1 and q_2 are separated by a centre to centre distance r. If they are made to touch each other and then separated to the same distance, the force between them will be
   a) less than before b) same as before
   c) more than before d) zero

2) A parallel plate capacitor stores a charge Q at a voltage V. Suppose the area of the parallel plate capacitor and the distance between the plates are each doubled then which is the quantity that will change?
   a) Capacitance b) Charge c) Voltage d) Energy density

3) A wire connected to a power supply of 230 V has power dissipation. P_1 Suppose the wire is cut into two equal pieces and connected parallel to the same power supply. In this case power dissipation is P_2. The ratio P_2 / P_1
   a) 1 b) 2 c) 3 d) 4

4) In India electricity is supplied for domestic use at 220 V. It is supplied at resistance of a 60W bulb for use in USA will be
   a) R b) 2R c) $\frac{R}{4}$ d) $\frac{R}{2}$

5) A thin insulated wire forms a plan spiral of N = 100 tight turns carrying a current I = 8mA (milli ampere). The radii of inside and outside turns area = 50 mm and b = 100 mm respectively. The magnetic induction at the centre of the spiral is
   a) 5 μT b) 7 μT c) 8 μT d) 10 μT

6) A flat dielectric disc of radius R carriesan excess charge on its surface. The surface charge density is σ. The discrotates about an axis perpendicular toits plane passing through the centre with angular velocity ω. Find the magnitude of the torque on the disc if it is placed in a uniform magnetic field whose strength is B which is directed perpendicular to the axis of rotation
   a) $\frac{1}{4}\sigma\omega\pi BR$ b) $\frac{1}{2}\sigma\omega\pi BR^2$ c) $\frac{1}{4}\sigma\omega\pi BR^3$ d) $\frac{1}{4}\sigma\omega\pi BR^4$

7) A circular coil of radius 5 cm and 50 turns carries a current of 3 ampere. The magnetic dipole moment of the coil is nearly
   a) 1.0 A m^2 b) 1.2 A m^2 c) 0.5 A m^2 d) 0.8 A m^2

8) A step-down transformer reduces the supply voltage from 220 V to 11 V and increase the current from 6 A to 100 A. Then its efficiency is
   a) 1.2 b) 0.83 c) 0.12 d) 0.9

9) In an oscillating LC circuit, the maximum charge on the capacitor is Q. The charge on the capacitor when the energy is stored equally between the electric and magnetic fields is
   a) $\frac{Q}{2}$ b) $\frac{Q}{\sqrt{3}}$ c) $\frac{Q}{\sqrt{2}}$ d) Q

10) The flux linked with a coil at any instant is given by $B = 10t^2 - 50t + 250$. The induced emf at t = 3s is
   a) −190 V b) −10 V c) 10 V d) 190 V

11) Which one of them is used to produce a propagating electromagnetic wave?
   a) an accelerating charge
   b) a charge moving with constant Velocity
   c) a stationary charge d) an uncharged particle

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12) The electric and magnetic fields of an electromagnetic wave are
a) in phase and perpendicular to eachother
b) out of phase and not perpendicular to eachother
c) in phase and not perpendicular to eachother
d) out of phase and perpendicular to eachother

13) An object is placed in front of a convex mirror of focal length of f and the maximum and minimum distance of an object from the mirror such that the image formed is real and magnified.
a) 2f and c b) c and ∞ c) f and O d) None of these

14) If the velocity and wavelength of light in air is Va and λa and that in water is Vw and λw, then the refractive index of water
a) $\frac{V_w}{V_a}$ b) $\frac{V_a}{V_w}$ c) $\frac{\lambda_w}{\lambda_a}$ d) $\frac{V_a\lambda_a}{V_w\lambda_w}$

15) Two coherent monochromatic lightbeams of intensities I and 4I are superposed. The maximum and minimum possible intensities in the resulting beam are
a) 5I and I b) 5I and 3 c) 9I and I d) 9I and 3I

II. Answer any six questions (Question no.24 is compulsory): 6×2=12

16) What is corona discharge?
17) State the principle of potentiometer.
18) State Fleming's left hand rule.
19) What are the limitations of the cyclotron?
20) A circular antenna of area 3 m^2 is installed at a place in Madurai. The plane of the area of antenna is inclined at 47° with the direction of Earth's magnetic field. If the magnitude of Earth's field at that place is 4.1×10^{-5} T find the magnetic flux linked with the antenna.
21) Mention the ways of producing induced emf.
22) What is displacement current?
23) Why does sky appear blue?
24) The wavelength of a light is 450 nm. How much phase it will differ for a path of 3 mm?

III. Answer any six questions: (Question no.33 is compulsory) 6×3=18

25) Derive the expression for resultant capacitance, when capacitors are connected in parallel.
26) Explain the determination of the internal resistance of a cell using voltmeter.
27) If the resistance of coil is 3 Ω at 20°C α = 0.004/°C then determine its resistance at 100°C.
28) Discuss the conversion of galvanometer into a voltmeter.
29) Explain Biot-Savart law.
30) How will you induce an emf by changing the area enclosed by the coil?
31) Write the uses Infra red rays.
32) Mention the differences between interference and diffraction.
33) Find the ratio of the intensities of lights with wavelengths 500 nm and 300 nm which undergo Rayleigh scattering.

IV. Answer all the questions: 5×5=25

34) a) Explain in detail the construction and working of a Van de Graaff generator. **(OR)**
b) Explain the determination of unknown resistance using meter bridge.

35) a) Discuss the working of cyclotron in detail. **(OR)**
b) Derive the expression for the force between two parallel, current-carrying conductors.

36) a) Explain the construction and working of transformer. **(OR)**
b) Derive an expression for phase angle between the applied voltage and current in a series RLC circuit.

37) a) Explain the types of absorption spectrum. **(OR)**
b) Obtain the equation for bandwidth in Young's double slit experiment.

38) a) Describe the Fizeau's method to determine the speed of light. **(OR)**
b) Obtain lens maker's formula and mention its significance.

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